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Poster Requirements and Guidelines

for

Mechanical Project 478
Mechatronic Project 478/488

January 2024

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1. INTRODUCTION

The purpose of the poster is to communicate the topic and results of the project to the general public or, in other words, someone that does **not** have an engineering background. The target audience for these posters is a high-school learner.

2. REQUIREMENTS

- 2.1. Poster size: A2 in landscape format.
- 2.2. Only English may be used (because that is the language used in engineering practice).
- 2.3. The following information must be given in logical order:
 - a. title (can be a shortened version of the project title),
 - b. student's name, followed by the supervisor's name,
 - c. date,
 - d. short description of the main objective(s), procedures, typical results and important conclusions,
 - e. recognition of any financial or physical support from an external institution.
- 2.4. Excluding the title, authors and date, no more than **200 words**, including captions of figures and illustrations, may be used on the poster.
- 2.5. The poster's text must be readable from a distance of at least 1 m.
- 2.6. Sketches, graphs, photo's, etc. must only show the important trends and must be readable from a distance of 1 m. Note that this affects not only the overall size of illustrations, but also the amount of detail in it.
- 2.7. The layout of the example in Figure 1 must be adhered to.

Powder Injection Moulding of Ti-6Al-4V Titanium Alloy: Feedstock and Mixer

Author: JE Hoffman Supervisor: Prof DC Blaine
Department of Mechanical and Mechatronic Engineering
January 2024

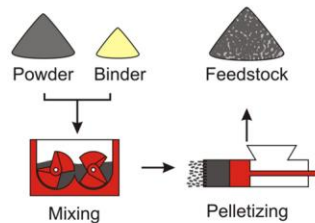
Introduction

Powder injection moulding (PIM) is a proven method to fabricate small parts of complex geometries. Further development of PIM technologies for use with titanium requires the creation of a suitable feedstock.

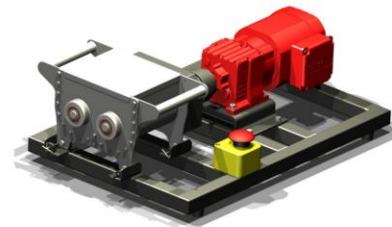
Objectives

The aim of this study is to create a feedstock for titanium that is suitable for PIM. To facilitate feedstock creation, a suitable mixer must be created.

Method



Mixing chamber of the designed sigma blade mixer



Mixer assembly

Results



The homogeneously mixed thermoplastic feedstock consists of gas atomised Ti-6Al-4V powder with a binder layer consisting of mixed paraffin wax, polyethylene, polypropylene and stearic acid.

Conclusions

The designed sigma blade mixer and above combination of materials proved successful in creating a homogeneous feedstock.

Acknowledgements

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Figure 1: Example of a poster

3. GENERAL GUIDELINES¹

3.1 An Effective Poster

- An effective poster is a **visual** communications tool.
- An effective poster is:
 - **Focused** – Focused on a single message.
 - **Graphic** – Let graphs and images tell the story; use text sparingly.
 - **Ordered** – Keep the sequence well-ordered and obvious.

3.2 Define Your Message

- Your goal is to convey a clear message and support it with a compelling combination of images and short blocks of text.
- Focus on your message throughout the poster. If it does not reinforce your message, leave it out!
- If you have an interesting result, state it explicitly in the title.
- Make the strongest statements your data will support.

3.3 Know Your Audience

There are three categories of readers in most audiences, Table 1:

- your field of specialization
- fields closely related to yours
- unrelated fields

To satisfy them all, you should:

- Provide context for your work. Explain the big picture and why the problem is important.
- Use plain language to present your work.
- Avoid jargon and acronyms unless you're *really* positive that yours will be a specialist-only audience.
- Interpret your findings so that readers in all categories can understand how your work helps solve the problem you've described.

¹ Hess, G.R., K. Tosney, and L. Liegel. 2006. Creating Effective Poster Presentations. URL=<http://www.ncsu.edu/project/posters>, visited 3 November 2006

Table 1: Audience categories

Audience Category	Requirements
People in your field of specialization , including your colleagues.	No special efforts are required to attract them. They will read whatever you present, no matter how well or poorly you present it.
People in fields closely related to yours are worth capturing, because they can have interesting insights and perspectives about your work.	They will require that you supply context for your work. They are likely to be unfamiliar with your jargon.
People in unrelated fields can be attracted by an accessible message, and provide valuable insights and links to distant fields.	They require you to explain the problem and the solution. They will not understand your jargon.

3.4. Create Your Poster

Each of the following aspects of poster creation is considered in detail.

- **Planning** - Before starting work on your poster, consider message, space, format (single sheet or multi-panel), and deadlines.
- **Focus** - Stay focused on your message and keep it simple. Create a mock-up and dispense with unneeded details.
- **Layout** - Use a clearly defined visual grammar to move readers through your poster.
- **Headings** - Use headings to orient readers and convey major points.
- **Graphics** - Clear graphics should dominate your poster.
- **Text** - Text should be minimized in favour of graphics, and large where used.
- **Colours** - Colours can make a poster attractive and improve readability, but be cautious.
- **Editing** - Edit ruthlessly to reduce the amount of text and focus on a results-oriented message.
- **Software** - There are many packages you can use to create your poster.

3.4.1 Planning

What's my message? You must be able to state your main points and conclusions clearly. **All** visuals and text should relate to those points and conclusions.

How much room do I have? Determine specific size requirements and consider layout (landscape vs. portrait orientation).

3.4.2 Focus

- Simple messages are more memorable.
- Details distract from the main point, and can be supplied in person as needed.
- Ask yourself which details are *absolutely essential* for conveying your message.
- Omit anything that is not essential.
- Edit text carefully - simplify verbiage, reduce sentence complexity.

3.4.3 Layout

Your poster should:

- **use a visual grammar** to guide readers to the important parts of your poster.
- **use a column format** to make your poster easier to read in a crowd.
- **use organization cues** to guide readers through your poster.
- **use "reader gravity"** which pulls the eye from top to bottom and left to right.
- **use headings intelligently** to help readers find your main points and key information.
- **balance the placement of text and graphics** to create visual appeal.
- **use white space creatively** to help define the flow of information.

Visual grammar is a graphic hierarchy that helps readers identify the most important parts of your poster. If it's important, make it BIG. Use type size proportional to importance, Figure 2. Use simple figures and graphs, which should dominate the poster visually. Make all graphic elements large enough to be visible easily from one meter away.

Use a columnar format, Figure 3. It allows readers to read the entire poster as they proceed from left to right. If the poster is organized in columns, viewers can read all of a column before they move to the next column. Readers tend to read top to bottom, a phenomenon called "reader gravity".

You can also use numbers, letters, or arrows to help guide viewers through your poster, Figure 4.

Your poster should have a good visual balance of figures and text, separated by white space. Balance occurs when images and text are reflected (at least approximately) across a central horizontal, vertical, or diagonal axis. This axis is known as the axis of symmetry, Figure 5.

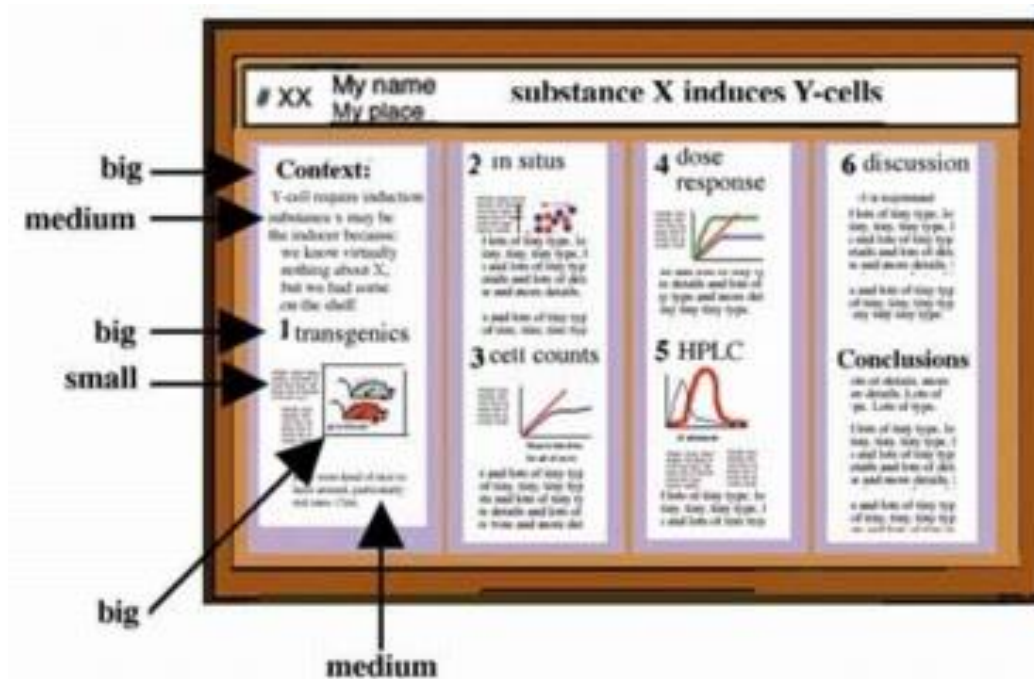


Figure 2: Font size

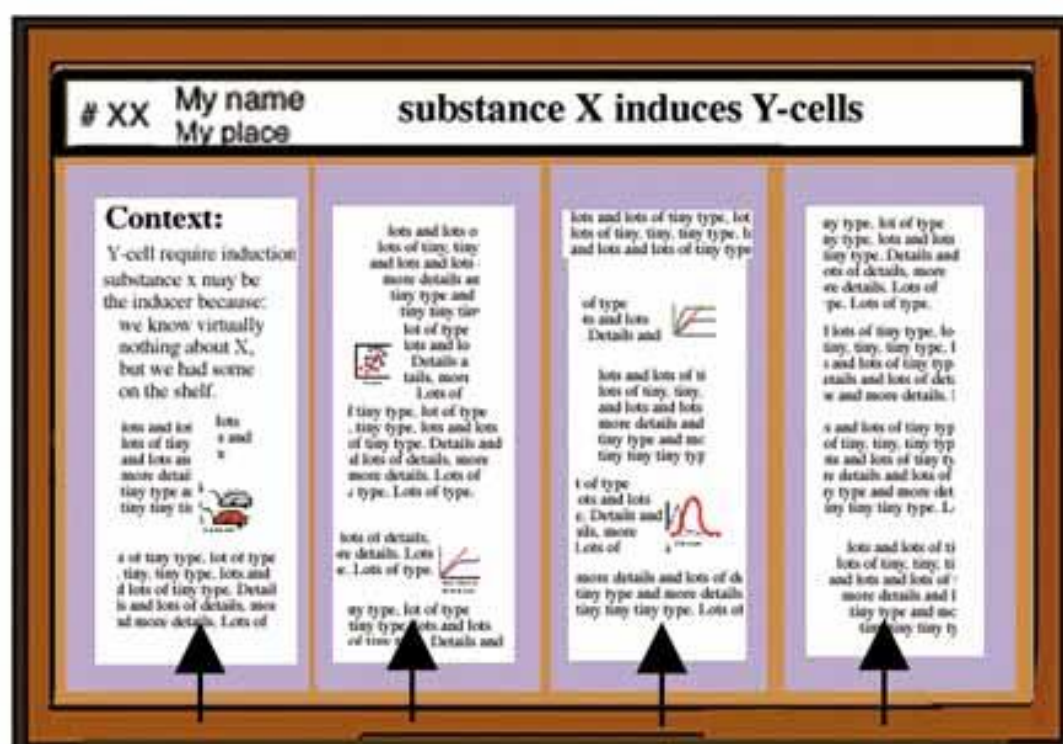


Figure 3: Column format

3.4.4 Headings

Use headings to orient readers and convey major points. Headings - including the title, section titles, and figure captions – should:

- **Summarize** - Use headings as opportunities to summarize your work in large letters. A hurried reader should be able to get the main points from the headings alone.
- **Organize** - Good headings are part of the visual grammar that helps move readers through your poster.
- **Be Hierarchical** - The more important the point, the larger the type.
- **Be Bold** - Make the strongest statements your research allows.

In the example in Figure 6, note how headings convey the message - viewers in a hurry need not read further.

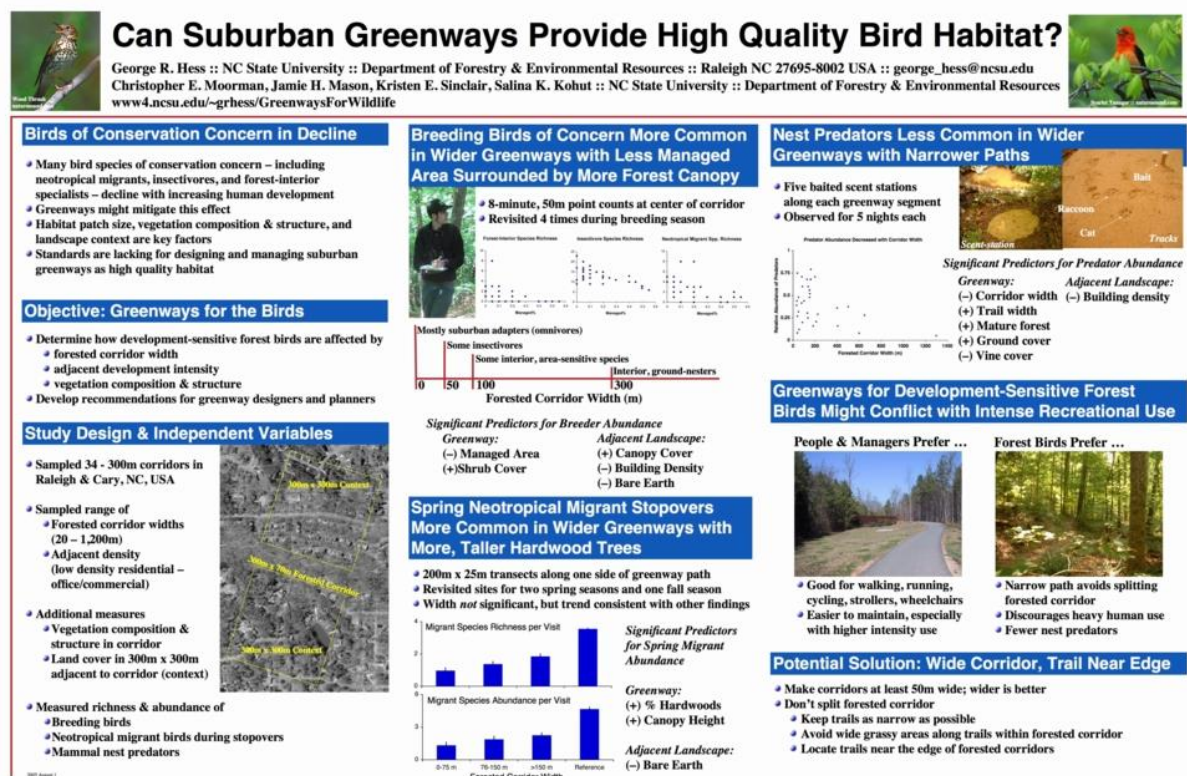


Figure 6: The use of heading to convey the most important aspects

3.4.5 Graphics

Simple, clean graphics communicate *relationships* quickly. Good graphics - graphs, illustrations, photos - are the centrepiece of your poster.

- Good graphs communicate relationships quickly.
- Graphs should be simple and clean.
- Write explanations directly on figures, instead of referencing from elsewhere.

- Use simple 2-dimensional line graphs, bar charts, pie charts.
- Avoid 3-dimensional graphs unless you're displaying 3-dimensional data - and then proceed carefully, as many 3-D graphs are difficult to interpret.
- Text on graphs must follow same guidelines as all other text so that it will be visible.
- Use photos that help deliver your message.
- Use graphics, cartoons, and figures instead of text.
- Use colour to emphasize or to link words and images together.
- Use bold lines and obvious pattern or colour to distinguish figures.
- Write explanations directly on the figures.
- Minimize abbreviations and cross-references.

Figure 7 shows a very simple example of the relationship between two populations through time. The main point to be made is that the lynx (predator) and hare (prey) populations oscillate through time in a somewhat predictable manner. Straight out of Excel, you will get Figure 7.

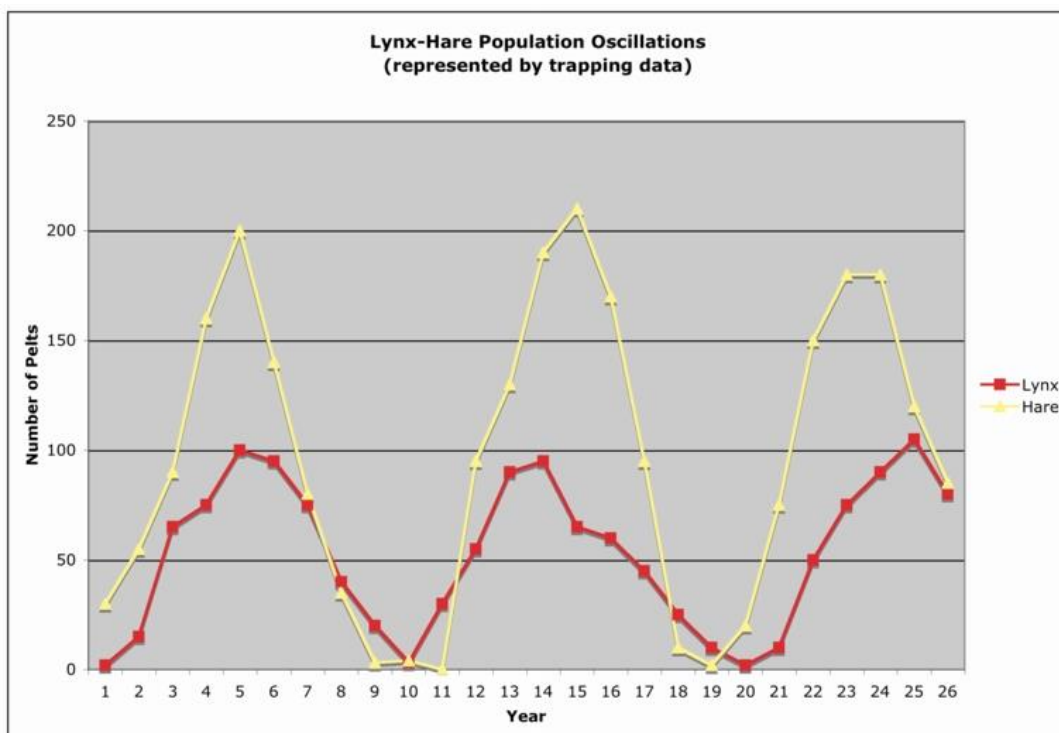


Figure 7: Standard Excel graph

There's a lot of ink here that doesn't convey information relevant to the main point you're trying to make. Plus it's ugly.

- **Grey background:** not only does it provide absolutely no information, it's also unsightly. After you remove it, you will likely have to darken some of the lines.

- **Grid lines:** it's very unlikely that your audience cares about the exact values at each data point - it's the pattern that matters. The grid lines compete with the pattern you're trying to show.
- **Legend:** it's taking up space that would be better spent on the graph
- **X-axis:** The labelling between tick marks is confusing.
- **Axis scales:** Because the pattern is the main focus, we don't need to have such finely detailed scales - just enough to provide a sense of the range of values plotted.
- **Y-axis label:** Why make the reader tilt his or her head to read?
- **Legend:** Why make the reader look back and forth between lines and legend? Just label the lines - then eliminate the legend.
- **Line types:** Use colour *and* line type to differentiate - this will help people who have colour impaired vision, and also any grey-scale copies of the poster you make (as for handouts).

It's easy to make a graph that looks cleaner and has a higher ratio of information-to-total ink as shown in Figure 8.

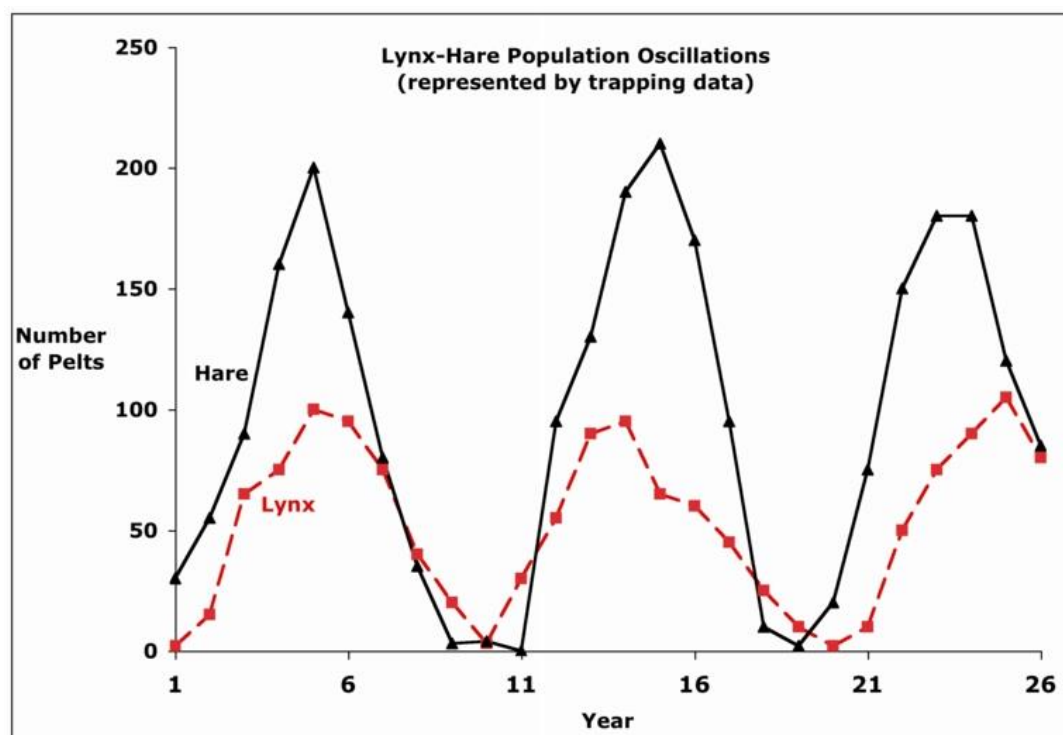


Figure 8: Improved graph from Excel

3.4.6 Text

Text should be simple, direct, and large.

- Minimize text - use images and graphs instead.
- Use phrases rather than full sentences.

- Use an active voice.
- Left-justify text; avoid centering and right-justifying text.
- Use a serif font (e.g., Times) for most text - easier to read.
- Sans-serif font (e.g., Helvetica) OK for titles and headings.
- Text should be **at least** 24 points in text, 36 for headings.
- Pay attention to text size in figures - it must also be large.
- Title should be **at least 5cm tall**.
- Title and major headings should be readable from 2m away. Headings carry your main message and can attract viewers to your poster.
- Supporting material should be visible from 1m away.
- Avoid long lines of text.
- Keep text blocks compact - single spaced and 50 words or fewer - and separate with white space.

Hint: If you print your complete poster on a single A4 sheet of paper, you should be able to read all of it - including text in figures - comfortably. If you can't, your text is too small.

3.4.7 Colours

Use colour to attract attention, organize, and emphasize - but don't overdo it.

- Use a light colour background and dark colour letters for contrast.
- Avoid dark backgrounds with light letters - very tiring to read.
- Stick to a theme of 2 or 3 colours - much more will overload and confuse viewers.
- If you use multiple colours, use them in a consistent pattern - otherwise viewers will spend their time wondering what the pattern is rather than reading your poster.
- Overly bright colours will attract attention - and then wear out readers' eyes.
- Consider people who have problems differentiating colours, especially when designing graphics - one of the most common is an inability to tell green from red.

3.4.8 Editing

If it doesn't provide critical support for your main message, eliminate it! Edit all text to reduce sentence complexity, and to delete details.

- If it's not relevant to your message, remove it!
- Have colleagues comment on drafts. Print a small version and circulate for comment, or hang a full-size draft with pens and invite them to critique.
- Are your objective and main message obvious?

3.4.9 Software

- Microsoft PowerPoint is a relatively easy-to-use tool for creating posters. One benefit of PowerPoint is that many people already know how to use it.
- Adobe Illustrator and InDesign have more features and can provide very professional results - especially for posters including lots of high-resolution images - but are more complex and expensive.
- Microsoft Excel can create graphics and export them for PowerPoint - but you'll need to clean them up. DeltaGraph is also a nice tool, but keep your graphs clean.
- Adobe Photoshop is great for manipulating images, but is also complex and expensive.